Diffusion Probabilistic Model

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Reference (1)

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Reference (2)

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Goal of a Generative Model

For an generative model, what we want to obtain is
given training samples $\mathbf{x}$,
we want to train the model to learn the true data distribution $p(\mathbf{x})$.

ELBO (1)

Our goal is to maximize the log likelihood $\log p(\mathbf{x})$, or the evidence, of all observed samples. But we are not able to directly maximize the evidence since we have no access to the true distribution.
Thus, we bound the term with ELBO as our objective. We need the term being able to be calculated w.r.t our model parameters $\boldsymbol{\phi}$
the lowerbound of our evidence (ELBO) is given as

$$\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right]$$

pf: ELBO is the lower bound of the evidance.

ELBO (2)

by conditional probability,

$$\log p(\mathbf{x}) = \log p(\mathbf{x}) \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) d\mathbf{z} \qquad \left[\int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) d\mathbf{z} = 1\right] \tag{1}$$

$$= \int \log p(\mathbf{x}) q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) d\mathbf{z} \tag{2}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \left[\log p(\mathbf{x})\right] \tag{3}$$

ELBO (3)

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\right] \quad [\text{by conditional probability}] \tag{4}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{p(\mathbf{z}|\mathbf{x})}\frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \quad \left[\text{multiply by } 1 = \frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \tag{5}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}{p(\mathbf{z}|\mathbf{x})}\right] \tag{6}$$

ELBO (4)

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] + D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}|\mathbf{x})) \tag{7}$$

$$\geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \tag{8}$$

We want to train our variational model to learn the true posterior of the latent variables

$$q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \rightarrow p(\mathbf{z}|\mathbf{x})$$

by minimizing the KL-divergence term

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p(\mathbf{z}|\mathbf{x}))$$

but since we don't know the true posterior $p(\mathbf{z}|\mathbf{x})$, we are not able to directly minimize the KL-divergence term.

ELBO (5)

But we also have a property that $\log p(\mathbf{x})$ is independent of our model $\boldsymbol{\phi}$, thus is a constant over all possible parameters $\boldsymbol{\phi}$. That is

$$\log p(\mathbf{x}) = \text{ELBO} + D_{\text{KL}} = \text{constant}$$

Therefore, by maximizing ELBO w.r.t $\boldsymbol{\phi}$, we are minimizing D_{KL}, forcing $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \sim p(\mathbf{z}|\mathbf{x})$,

Variational Autoencoders (1)

From [3], a VAE aims to iteratively learn the encoder and decoder. In other words, we try to make the two distribution, the true posterior distribution $p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})$, and the approximate posterior $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, be as close as possible.

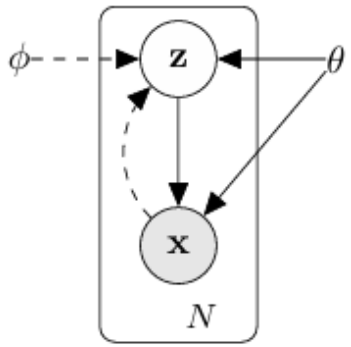


Figure 1: The type of directed graphical model under consideration. Solid lines denote the generative model $p_{\boldsymbol{\theta}}(\mathbf{z})p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})$, dashed lines denote the variational approximation $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ to the intractable posterior $p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})$. The variational parameters $\boldsymbol{\phi}$ are learned jointly with the generative model parameters $\boldsymbol{\theta}$.

Variational Autoencoders (2)

If we further expand the ELBO, and also notice how the "true distribution" $p(\cdot)$ is parameterized as an encoder $p_{\boldsymbol{\theta}}(\cdot)$

$$\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p_{\boldsymbol{\theta}}(\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \quad \text{[conditional prob.]} \tag{9}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right] + \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] \tag{10}$$

$$= \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right] - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p_{\boldsymbol{\theta}}(\mathbf{z})) \tag{11}$$

Variational Autoencoders (3)

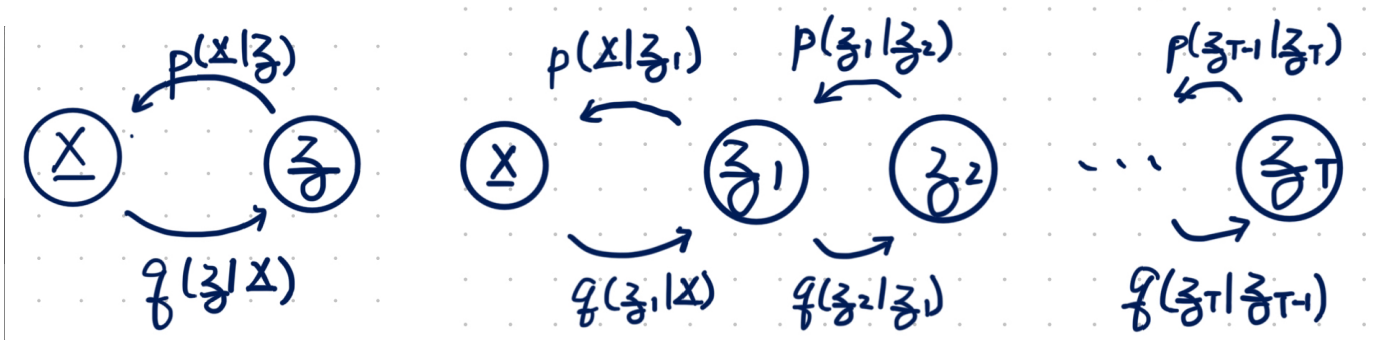
$$\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x},\mathbf{z})}{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\right] = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right] - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p_{\boldsymbol{\theta}}(\mathbf{z})) \tag{12}$$

From the expansion on the ELBO, we can now see that, by maximizing ELBO, we are

- maximizing $\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})}\left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})\right]$ the log likelihood of the decoder $p_{\boldsymbol{\theta}}$
- minimizing $D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})||p_{\boldsymbol{\theta}}(\mathbf{z}))$ the distance between the encoder $q_{\boldsymbol{\phi}}$ and the decoder $p_{\boldsymbol{\theta}}$

Hierarchical Variational Autoencoders (1)

the latent variable of VAE is extended to multiple hierarchies T to form a HVAE, where each level of the latent variables are modeled by a higher level latents.



(a) VAE (b) HVAE

Hierarchical Variational Autoencoders (2)

In the paper [5] [4] they usually assume the transition along hierarchy is Markovian.
The joint distribution of data $\mathbf{x}$ and all latents $\mathbf{z}_{1:T}$ can now be written as

$$p(\mathbf{x}, \mathbf{z}_{1:T}) = p(\mathbf{z}_T) p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_1) \prod_{t=2}^{T} p_{\boldsymbol{\theta}}(\mathbf{z}_{t-1}|\mathbf{z}_t),$$

and the posterior is

$$q_{\boldsymbol{\phi}}(\mathbf{z}_{1:T}|\mathbf{x}) = q_{\boldsymbol{\phi}}(\mathbf{z}_1|\mathbf{x}) \prod_{t=2}^{T} q_{\boldsymbol{\phi}}(\mathbf{z}_t|\mathbf{x}_{t-1})$$

Variational Diffusion Models (1)

The VDM can be treated as an HVAE with the following constraints:

- The latents $\mathbf{z}_t$ has exactly the same dimention as the data $\mathbf{x}$, therefore denoted as $\mathbf{x}_t$
- the encoders $q(\mathbf{x}_{t+1}|\mathbf{x}_t)$ is not modeled by the parameters, but pre-determined by a Gaussian distribution
  $q(\mathbf{x}_t|\mathbf{x}_{t-1}) \sim \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$
- the latent at timestep $\mathbf{x}_T$ is designed to be a standard Gaussian.
  $p(\mathbf{x}_T) \sim \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$

Variational Diffusion Models (2)

We therefore have the posterior, exactly like in HVAE, with some re-indexing,

$$q_{\boldsymbol{\phi}}(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q_{\boldsymbol{\phi}}(\mathbf{x}_t|\mathbf{x}_{t-1})$$

and the joint distribution

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^{T} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

The below are some comments about the α scaling.
Notice the distribution can be written as

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) \sim \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I}) \tag{13}$$

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \tag{14}$$

Variational Diffusion Models (3)

where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
This scaling maintains the variance of $\mathbf{x}_t$ across different timesteps. If we assume $\text{Var}(x_{t-1}) = 1$, and assume the noise and the signals are independent.

$$\text{Var}(\mathbf{x}_t) = \alpha_t \,\text{Var}(\mathbf{x}_{t-1}) + (1 - \alpha_t)\,\text{Var}(\boldsymbol{\epsilon}_t) \tag{15}$$

$$= \alpha_t + 1 - \alpha_t = 1 \tag{16}$$

Thus, if we always normalize the input signals x_0 to have a unit variance, all $\mathbf{x}_t$ will always have a unit variance with this defined variance schedule. This normalization ensures that the variance does not vanish or explode, and maintain the balance of signal and noise. And if we make $\{\alpha_t\}$ monotonically decreasing, $\{\bar{\alpha}_t\} \to 0$ and the last timestep

$$\mathbf{x}_T = \sqrt{\bar{\alpha}_T}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon_T \simeq \epsilon_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

ELBO of VDM (1)

Because we have the Markovian assumption. Then we expand it with Bayse rule

$$q(\mathbf{x}_t|\mathbf{x}_{t-1},\mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1},\mathbf{x}_0|\mathbf{x}_t)q(\mathbf{x}_t)}{q(\mathbf{x}_{t-1},\mathbf{x}_0)} \quad \text{(Bayes Rule)} \tag{17}$$

$$= \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0,\mathbf{x}_t)q(\mathbf{x}_0|\mathbf{x}_t)q(\mathbf{x}_t)}{q(\mathbf{x}_{t-1},\mathbf{x}_0)} \quad \text{(Bayes Rule)} \tag{18}$$

$$= \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0,\mathbf{x}_t)q(\mathbf{x}_0,\mathbf{x}_t)}{q(\mathbf{x}_{t-1},\mathbf{x}_0)} \quad \text{(Conditional Probability)} \tag{19}$$

$$= \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0,\mathbf{x}_t)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \quad \text{(Conditional Probability)} \tag{20}$$

Starting from ELBO,

ELBO of VDM (2)

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}_0) \tag{21}$$

$$\geq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \tag{22}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \tag{23}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0)}\right] \quad \text{(Markov Chain)} \tag{24}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{p(\mathbf{x}_T)p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}}\right] \quad \text{(Bayes Rule)} \tag{25}$$

ELBO of VDM (3)

Notice for the product

$$\prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} = \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

We now focus on the reduction of the second fraction

$$\prod_{t=2}^{T} \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} = \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{\cancel{q(\mathbf{x}_2|\mathbf{x}_0)}} \frac{\cancel{q(\mathbf{x}_2|\mathbf{x}_0)}}{\cancel{q(\mathbf{x}_3|\mathbf{x}_0)}} \cdots \frac{\cancel{q(\mathbf{x}_{T-2}|\mathbf{x}_0)}}{\cancel{q(\mathbf{x}_{T-1}|\mathbf{x}_0)}} \frac{\cancel{q(\mathbf{x}_{T-1}|\mathbf{x}_0)}}{q(\mathbf{x}_T|\mathbf{x}_0)} = \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)}$$

therefore

$$\prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}} = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}$$

ELBO of VDM (4)

the expectation can now be written as

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_1|\mathbf{x}_0)} \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{q(\mathbf{x}_T|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \tag{26}$$

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}} \frac{\cancel{q(\mathbf{x}_1|\mathbf{x}_0)}}{q(\mathbf{x}_T|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \tag{27}$$

ELBO of VDM (5)

$$= \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T) p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)}{q(\mathbf{x}_T|\mathbf{x}_0)} \prod_{t=2}^{T} \frac{p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \tag{28}$$

$$\begin{aligned} = \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log(p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)) \right] + \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} \right] \\ + \sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] \end{aligned} \tag{29}$$

we want to marginalize the latents at other timesteps

ELBO of VDM (6)

if we take out one term and expand

$$\mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\left[\log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)}\right] = \int q(\mathbf{x}_{1:T}|\mathbf{x}_0) \log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} d\mathbf{x}_{1:T} \tag{30}$$

$$= \int \int q(\mathbf{x}_{1:T}|\mathbf{x}_0) d\mathbf{x}_{1:T-1} \log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} d\mathbf{x}_T \tag{31}$$

$$= \int q(\mathbf{x}_T|\mathbf{x}_0) \log \frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)} d\mathbf{x}_T = -D_{\mathrm{KL}}\left[q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T)\right] \tag{32}$$

$$\tag{33}$$

Similarly, we marginalize the latents at each term

ELBO of VDM (7)

$$= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\left[\log(p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1))\right] + \mathbb{E}_{q(\mathbf{x}_T|\mathbf{x}_0)}\left[\log\frac{p(\mathbf{x}_T)}{q(\mathbf{x}_T|\mathbf{x}_0)}\right] \tag{34}$$

$$+ \sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_{t-1}|\mathbf{x}_0)}\left[\log\frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right] \tag{35}$$

we try to rewrite the sum

$$\sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t,\mathbf{x}_{t-1}|\mathbf{x}_0)}\left[\log\frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}\right] \tag{36}$$

$$\tag{37}$$

ELBO of VDM (8)

expand the expectation we will get

$$\sum_{t=2}^{T} \int \int q(\mathbf{x}_t, \mathbf{x}_{t-1}|\mathbf{x}_0) \log \frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} d\mathbf{x}_t d\mathbf{x}_{t-1} \tag{38}$$

$$= \sum_{t=2}^{T} \int \int q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) q(\mathbf{x}_t|\mathbf{x}_0) \log \frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} d\mathbf{x}_t d\mathbf{x}_{t-1} \tag{39}$$

$$= \sum_{t=2}^{T} \int q(\mathbf{x}_t|\mathbf{x}_0) \int q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \log \frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} d\mathbf{x}_{t-1} d\mathbf{x}_t \tag{40}$$

$$= \sum_{t=2}^{T} \int q(\mathbf{x}_t|\mathbf{x}_0) \mathbb{E}_{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \left[\log \frac{p(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)} \right] d\mathbf{x}_t \tag{41}$$

$$\tag{42}$$

ELBO of VDM (9)

$$= -\sum_{t=2}^{T} \int q(\mathbf{x}_t|\mathbf{x}_0) D_{\mathrm{KL}} \left[\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] d\mathbf{x}_t \tag{43}$$

$$= -\sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[D_{\mathrm{KL}} (\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \tag{44}$$

$$\tag{45}$$

Therefore the ELBO can now be interpreted as

$$\begin{aligned} &= \mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} \left[\log(p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1)) \right] - D_{\mathrm{KL}} \left[q(\mathbf{x}_T|\mathbf{x}_0) || p(\mathbf{x}_T) \right] \\ &\quad - \sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[D_{\mathrm{KL}} (q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) || p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \right] \end{aligned} \tag{46}$$

ELBO of VDM (10)

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \text{ELBO} = \underbrace{\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}\left[\log(p_{\boldsymbol{\theta}}(\mathbf{x}_0|\mathbf{x}_1))\right]}_{L_0} \tag{47}$$

$$- \underbrace{D_{\text{KL}}\left[q(\mathbf{x}_T|\mathbf{x}_0)||p(\mathbf{x}_T)\right]}_{L_T} \tag{48}$$

$$- \sum_{t=2}^{T} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \underbrace{\left[D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))\right]}_{L_{t-1}} \tag{49}$$

- L_0: The log-likelihood of the last reconstruction step
- L_T: the KL divergence at the last timestep.
- L_{t-1}: matching the denoising steps ($p_{\boldsymbol{\theta}}$) to the ground truth (q) at each timesteps. Now can be computed with only one random variable at a time.

Further Expansion with Gaussian Assumption

By the above we only utilized the Markov chain assumption. we want to further simplify $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ in L_{t-1} term. Notice this term is

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

and we already have

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I})$$

from reparameterization we can write

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \text{ with } \boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\boldsymbol{\epsilon}; \mathbf{0}, \mathbf{I})$$

Thus the following goal is to rewrite $q(\mathbf{x}_t|\mathbf{x}_0)$ into a Gaussian distribution.

$\mathbf{x}_t$ (1)

From reparameterization we can write $\mathbf{x}_t \sim q(\mathbf{x}_t|\mathbf{x}_{t-1})$ the latent with

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}$$

recursively substitute each step of latent

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \\
&= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2}) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}
\end{aligned}$$

Here since $\boldsymbol{\epsilon}$ are sampled from a standard Gaussian Distribution, we treat

$$\sqrt{\alpha_t - \alpha_{t-1}}\boldsymbol{\epsilon}_{t-2} \sim \mathcal{N}(\mathbf{0}, (\alpha_t - \alpha_{t-1})\mathbf{I})$$

and

$\mathbf{x}_t$ (2)

$$\sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, (1-\alpha_t)\mathbf{I})$$

therefore the sum of two independent gaussian can be treated as

$$(\sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1}) \sim \mathcal{N}(\mathbf{0}, ((\alpha_t - \alpha_t\alpha_{t-1}) + (1-\alpha_t))\mathbf{I})$$

thus we rewrite the sum as

$$\sqrt{\alpha_t - \alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} = \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}^*_{t-2}$$

with $\boldsymbol{\epsilon}^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

$\mathbf{x}_t$ (3)

therefore we can now keep expand $\mathbf{x}_t$ as

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t \alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}^*_{t-2} \\
&= \dots \\
&= \sqrt{\prod_{i=1}^{t}\alpha_i}\mathbf{x}_0 + \sqrt{1-\prod_{i=1}^{t}\alpha_i}\boldsymbol{\epsilon}^*_0 \\
&= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}^*_0 \\
&\sim \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, 1-\bar{\alpha}_t\mathbf{I})
\end{aligned}$$

so we now have

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, 1-\bar{\alpha}_t\mathbf{I})$$

$q(\mathbf{x}_t|\mathbf{x}_0)$ (1)

with the above expansion we can now have

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$
$$= \frac{\mathcal{N}(\mathbf{x}_t;\sqrt{\alpha_t}\mathbf{x}_{t-1},(1-\alpha_t)\mathbf{I})\mathcal{N}(\mathbf{x}_{t-1};\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0,(1-\bar{\alpha}_{t-1})\mathbf{I})}{\mathcal{N}(\mathbf{x}_t;\sqrt{\bar{\alpha}_t}\mathbf{x}_0,(1-\bar{\alpha}_t)\mathbf{I})}$$
$$\propto \exp\{-\frac{(\mathbf{x}_t-\sqrt{\alpha_t}\mathbf{x}_{t-1})^2}{2(1-\alpha_t)} - \frac{(\mathbf{x}_{t-1}-\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0)^2}{2(1-\bar{\alpha}_{t-1})} + \frac{(\mathbf{x}_t-\sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{2(1-\bar{\alpha}_t)}\}$$

We omit the scaling $\frac{1}{\sqrt{2\pi}\sigma}$ here for simplicity.
Expand the square in each term we have

$$= \exp\{-\frac{1}{2}\left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2}{(1-\alpha_t)} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1} + \bar{\alpha}_{t-1}\mathbf{x}_0^2}{(1-\bar{\alpha}_{t-1})} - \frac{(\mathbf{x}_t-\sqrt{\bar{\alpha}_t}\mathbf{x}_0)^2}{(1-\bar{\alpha}_t)}\right)$$

$q(\mathbf{x}_t|\mathbf{x}_0)$ (2)

since $\mathbf{x}_t, \mathbf{x}_0$ are conditioned and all α are given, the terms in red can be seen as constants and therefore just scaling the exponential, we write

$$\propto \exp\{-\frac{1}{2}\left(\frac{-2\sqrt{\alpha_t}\mathbf{x}_t\mathbf{x}_{t-1} + \alpha_t\mathbf{x}_{t-1}^2}{(1-\alpha_t)} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0\mathbf{x}_{t-1}}{(1-\bar{\alpha}_{t-1})}\right)\}$$

(reorganize the equation with $\mathbf{x}_{t-1}$,)

$$= \exp\{-\frac{1}{2}\left(\mathbf{x}_{t-1}^2\left[\frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right] - 2\mathbf{x}_{t-1}\left(\frac{\sqrt{\alpha_t}}{1-\alpha_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0\right)\right)\}$$

(take out the scalar of $\mathbf{x}_{t-1}^2$,)

$$= \exp\{-\frac{1}{2}\left(\frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\bar{\alpha}_{t-1}}\right)\left(\mathbf{x}_{t-1}^2 - 2\mathbf{x}_{t-1}\frac{\frac{\sqrt{\alpha_t}}{1-\alpha_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0}{\frac{\alpha_t}{1-\alpha_t} + \frac{1}{1-\bar{\alpha}_{t-1}}}\right)\}$$

$q(\mathbf{x}_t|\mathbf{x}_0)$ (3)

$$= \exp\{-\frac{1}{2}\frac{1}{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1}^2 - 2\mathbf{x}_{t-1}\frac{(\frac{\sqrt{\alpha_t}}{1-\alpha_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1-\bar{\alpha}_{t-1}}\mathbf{x}_0)(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}})\}$$

$$= \exp\{-\frac{1}{2}\frac{1}{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1}^2 - 2\mathbf{x}_{t-1}\frac{(\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0)}{1-\bar{\alpha}_{t-1}})\}$$

$$\propto \exp\{-\frac{1}{2}\frac{1}{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}}}(\mathbf{x}_{t-1}^2 - 2\mathbf{x}_{t-1}\frac{(\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0)}{1-\bar{\alpha}_{t-1}}$$

$$+(\frac{(\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0)}{1-\bar{\alpha}_{t-1}})^2\ )\}$$

$$= \exp\{-\frac{(\mathbf{x}_{t-1} - \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_{t-1}})^2}{2\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}}}\}$$

$$\sim \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_{t-1}}}_{\mu_q(\mathbf{x}_t,\mathbf{x}_0)}, \underbrace{\frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_{t-1}}\mathbf{I}}_{\mathbf{\Sigma}_q(t)=\sigma_q^2(t)\mathbf{I}})$$

Objective with Gaussian Assumption (1)

to fit the model better into predicting the noise distribution, we design our model $p_{\boldsymbol{\theta}}$ as a Gaussian.

- for $\boldsymbol{\Sigma}_{\boldsymbol{\theta}}$, since variance in forward process q is given, we set $\boldsymbol{\Sigma}_{\boldsymbol{\theta}}(t) = \boldsymbol{\Sigma}_q(t)$
- we parameterize the mean with $\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)$

since now the forward q and the reverse $p_{\boldsymbol{\theta}}$ are assumed to be Gaussian, our L_{t-1} term

$$D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t))$$

can be simplified with the equation

$$D_{\mathrm{KL}}(\mathcal{N}(\mathbf{x}_1; \boldsymbol{\mu}_{x_1}, \boldsymbol{\Sigma}_{x_1})||\mathcal{N}(\mathbf{x}_2; \boldsymbol{\mu}_{x_2}, \boldsymbol{\Sigma}_{x_2}))$$

$$= \frac{1}{2}\left[\log\frac{|\boldsymbol{\Sigma}_{x_2}|}{|\boldsymbol{\Sigma}_{x_1}|} - \dim(\mathbf{x}) + \mathrm{tr}(\boldsymbol{\Sigma}_{x_2}^{-1}\boldsymbol{\Sigma}_{x_1}) + (\boldsymbol{\mu}_{\mathbf{x}_2} - \boldsymbol{\mu}_{\mathbf{x}_1})^T\boldsymbol{\Sigma}_{x_2}^{-1}(\boldsymbol{\mu}_{\mathbf{x}_2} - \boldsymbol{\mu}_{\mathbf{x}_1})\right]$$

Objective with Gaussian Assumption (2)

$$
\begin{aligned}
&\arg\min_{\boldsymbol{\theta}} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\
&= \arg\min_{\boldsymbol{\theta}} D_{\mathrm{KL}}(\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_q,\boldsymbol{\Sigma}_q(t))||\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_{\boldsymbol{\theta}},\boldsymbol{\Sigma}_q(t))) \\
&= \arg\min_{\boldsymbol{\theta}} \frac{1}{2}\left[\log\frac{|\boldsymbol{\Sigma}_q(t)|}{|\boldsymbol{\Sigma}_q(t)|} - \dim(\mathbf{x}) + \mathrm{tr}(\boldsymbol{\Sigma}_q^{-1}(t)\boldsymbol{\Sigma}_q(t)) + (\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)^T\boldsymbol{\Sigma}_q^{-1}(t)(\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)\right] \\
&= \arg\min_{\boldsymbol{\theta}} \frac{1}{2}\left[0 - \dim(\mathbf{x}) + \dim(\mathbf{x}) + (\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)^T\boldsymbol{\Sigma}_q^{-1}(t)(\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)\right] \\
&= \arg\min_{\boldsymbol{\theta}} \frac{1}{2}\left[(\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)^T(\sigma_q^2(t)\mathbf{I})^{-1}(\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)\right] \\
&= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}\left[||(\boldsymbol{\mu}_{\boldsymbol{\theta}}-\boldsymbol{\mu}_q)||^2\right]
\end{aligned}
$$

Objective with Gaussian Assumption (3)

we know from derivation,

$$\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_{t-1}}$$

therefore we set our model as

$$\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)}{1-\bar{\alpha}_{t-1}}$$

Objective with Gaussian Assumption (4)

our objective can be further simplified with

$$\arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}\left[||(\boldsymbol{\mu}_{\boldsymbol{\theta}} - \boldsymbol{\mu}_q)||_2^2\right]$$

$$= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}\left[|| \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)}{1-\bar{\alpha}_{t-1}} - \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_{t-1}} ||_2^2\right]$$

$$= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}\left[|| \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)}{1-\bar{\alpha}_{t-1}} - \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0}{1-\bar{\alpha}_{t-1}} ||_2^2\right]$$

$$= \arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)} \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_{t-1}}\left[||\hat{\mathbf{x}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) - \mathbf{x}_0||_2^2\right]$$

Objective with Gaussian Assumption (5)

another interpretation is by exploiting

$$\mathbf{x}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}$$

then the mean $\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0)$ can be further expanded by

$$\begin{aligned}
\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0) &= \frac{1}{1-\bar{\alpha}_t}[\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\mathbf{x}_0] \\
&= \frac{1}{1-\bar{\alpha}_t}[\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)\frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_0}{\sqrt{\bar{\alpha}_t}}] \\
&= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})\mathbf{x}_t}{1-\bar{\alpha}_t} + \frac{(1-\alpha_{t-1})\mathbf{x}_t}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}} + \frac{(1-\alpha_{t-1}\sqrt{1-\bar{\alpha}_t})\boldsymbol{\epsilon}_0}{(1-\bar{\alpha}_t)\sqrt{\alpha_t}}
\end{aligned}$$

Objective with Gaussian Assumption (6)

and simplified to

$$\boldsymbol{\mu}_q(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_0$$

thus we further parameterize our model as

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t, t)$$

Objective with Gaussian Assumption (7)

L_{t-1} now become

$$\begin{aligned}
&\arg\min_{\boldsymbol{\theta}} D_{\mathrm{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)||p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)) \\
=&\arg\min_{\boldsymbol{\theta}} D_{\mathrm{KL}}(\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_q,\boldsymbol{\Sigma}_q(t))||\mathcal{N}(\mathbf{x}_{t-1};\boldsymbol{\mu}_{\boldsymbol{\theta}},\boldsymbol{\Sigma}_q(t))) \\
=&\arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}[\|\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t,t) \\
&\qquad -\frac{1}{\sqrt{\alpha_t}}\mathbf{x}_t+\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}\sqrt{\alpha_t}}\boldsymbol{\epsilon}_0\|_2^2] \\
&\qquad =\arg\min_{\boldsymbol{\theta}} \frac{1}{2\sigma_q^2(t)}\frac{(1-\alpha_t)^2}{(1-\alpha_t)\alpha_t}[\|\boldsymbol{\epsilon}_0-\hat{\boldsymbol{\epsilon}}_\theta(\mathbf{x}_t,t)\|_2^2]
\end{aligned}$$

in DDIM they discard the constant scalar of the loss function, using MSE of the predicted noise as the loss function.

VDM

after training, the model learned the distribution, and

- by sampling Gaussian Noise from $p(\mathbf{x}_T)$ and run through $\mathbf{p}_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ we can have a new $\mathbf{x}_0$ which follows the original data distribution $p(\mathbf{x})$

Nonnegativity of KL-Divergence

i.e. $D_{KL}(p||q) \geq 0$

$$-D_{KL}(p||q) = -\sum_x p(x) \ln \frac{p(x)}{q(x)} \tag{50}$$

$$= \sum_x p(x) \ln \frac{q(x)}{p(x)} \tag{51}$$

$$\leq \sum_x p(x) \frac{q(x)}{p(x)} - 1 \quad (\ln(a) \leq a - 1 \ \forall a > 0) \tag{52}$$

$$= \sum_x q(x) - \sum_x p(x) \tag{53}$$

$$= 1 - 1 = 0 \quad \square \tag{54}$$

Jensen's Inequality (1)

for X an integrable random rariable.
and let $g : \mathbb{R} \to \mathbb{R}$ be a convex function
and let $Y = g(X)$ being also integrable.
Jensen's inequality holds

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$$

pf:
for a convex function $f(x)$ we know by it's definition,

$$f(x) \geq (x_0) + b(x - x_0), \ \forall x$$

for any point x_0 the graph lies above its tangent at the point. b here is the slope of the tangent at x_0.

Jensen's Inequality (2)

for our function of random variable g here we have

$$g(X) \geq g(\mathbb{E}[X]) + b(X - \mathbb{E}[X])$$

$$\begin{aligned}
\mathbb{E}[g(X)] &\geq \mathbb{E}[g(\mathbb{E}[X]) + b(X - \mathbb{E}[X])] && (55)\\
&= \mathbb{E}[g(\mathbb{E}[X])] + b\mathbb{E}[X - \mathbb{E}[X]] && (56)\\
&\text{(expectation being an linear operator)} && (57)\\
&= g(\mathbb{E}[X]) + b(\mathbb{E}[X] - \mathbb{E}[X]) && (58)\\
&= g(\mathbb{E}[X]) \quad \square && (59)
\end{aligned}$$